SBU Comps Complex Analysis Problems

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Problem 1 (2026 J-I-3)**.** Prove that for every integer $n \geq 0$ and every entire function f on the complex plane, f is a degree-n polynomial if and only if

$$\lim_{z \to \infty} \frac{zf'(z)}{f(z)} = n.$$

Solution. If f is a degree n polynomial then $zf'(z)$ is also a degree n polynomial whose leading coefficient is n times the leading coefficient of f (zero in the case where $n = 0$), so obviously

$$\lim_{z \to \infty} \frac{zf'(z)}{f(z)} = n.$$

Suppose now that the limit condition holds. The nonzero poles of $\frac{zf'(z)}{f(z)}$ are exactly the nonzero zeroes of $f(z)$, so in order for the given limit to be finite, the set of zeroes must be bounded (since the magnitude of a meromorphic function near a pole is unbounded). Thus $f^{-1}(0)$ is closed in $\mathbb{C}$ and bounded, hence compact, and it is discrete since f is obviously not the zero function; this means $f^{-1}(0)$ is finite.

Suppose that the zeroes of f, with multiplicity, are $z_1, z_2, \ldots, z_m$. Then, $f(z) = (z - z_1) \cdots (z - z_m)g(z)$ for a nowhere vanishing entire function g, which means that

$$\frac{zf'(z)}{f(z)} = \frac{z}{z - z_1} + \frac{z}{z - z_2} + \cdots + \frac{z}{z - z_m} + \frac{zg'(z)}{g(z)}.$$

Since g is nowhere zero we can write $g(z) = e^{h(z)}$ for some entire function h. Taking the limit gives

$$n = \lim_{z \to \infty} \frac{zf'(z)}{f(z)} = m + \lim_{z \to \infty} \frac{zg'(z)}{g(z)} = m + \lim_{z \to \infty} zh'(z),$$

so $zh'(z)$ is bounded and entire, hence $zh'(z) = 0 \cdot h'(0) = 0$ for all $z \in \mathbb{C}$. This implies the limit in the right-hand side is zero, i.e. $m = n$, and h and g are constant functions, say $g(z) = c$ for all z and nonzero c (if c were zero then the given limit would not make sense), so we have shown that $f(z) = c(z - z_1) \cdots (z - z_n)$. □

Problem 2 (2026 J-II-4)**.** Let U be an open subset of $\mathbb{C}$. Let $(f_j(z))$ be a collection of holomorphic functions on U that converges uniformly on every compact subset. Carefully prove that the pointwise limit is holomorphic.

Solution. Is this not just showing that f is holomorphic around any $z_0 \in U$ by Morera's theorem? □

Problem 3 (2025 A-I-3)**.** Among all monic polynomials of degree n, show that z^n minimizes

$$\sup_{|z|<1} |p(z)|.$$

Solution. The maximum magnitude of $p(z)$ on the interior of the unit disk is equal to the maximum along the disk's boundary, by continuity and the maximum modulus principle. Write $p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$ for $a_0, \ldots, a_n \in \mathbb{C}$ and $a_n = 1$. Then $|p(z)|^2$ can be written for z on the unit circle as

$$p(z) \cdot \overline{p(z)} = \sum_{j=0}^{n} \sum_{k=0}^{n} a_j \overline{a_k} \cdot z^{j-k},$$

so the average of $|p(z)|^2$ on the unit circle is

$$\frac{1}{2\pi} \int_0^{2\pi} p(e^{i\theta}) \cdot \overline{p(e^{i\theta})} \, \mathrm{d}\theta = \frac{1}{2\pi} \sum_{0 \le j,k \le n} a_j \overline{a_k} \int_0^{2\pi} e^{i\theta(j-k)} \, \mathrm{d}\theta.$$

The integral of $e^{i\theta(j-k)}$ is zero when $j \neq k$ because the real and imaginary parts are $\cos((j-k)\theta)$ and $\sin((j-k)\theta)$, both of which integrate to zero over their period when $j - k \neq 0$, and 2π when $j = k$, since in that case the integrand is just 1. That is, the average of $|p(z)|^2$ on the unit circle is

$$\frac{1}{2\pi} \int_0^{2\pi} p(e^{i\theta}) \cdot \overline{p(e^{i\theta})} \, \mathrm{d}\theta = \frac{1}{2\pi} \sum_{j=0}^{n} a_j \overline{a}_j = 1 + |a_{n-1}|^2 + \cdots + |a_1|^2 + |a_0|^2 \ge 1,$$

so for any monic polynomial, $\sup_{|z|<1} |p(z)|$ is at least 1, with equality (corresponding to the case where almost every point on the boundary has magnitude 1) if and only if each of $a_0, \ldots, a_{n-1}$ are 0, i.e. $p(z) = z^n$. □

Problem 4 (2025 A-II-3)**.** Let $f : \mathbb{D} \to U$ be a conformal map from the unit disk $\mathbb{D} = \{z : |z| < 1\} \subset \mathbb{C}$ to a domain $U \subset \mathbb{C}$ of bounded area. For $\theta \in [0, 2\pi)$, let $R_\theta = \{re^{i\theta} : 1/2 \le r < 1\}$. Show that

$$\int_0^{2\pi} \mathrm{length}(f(R_\theta)) \, d\theta < \infty.$$

Show that for almost any angle θ, there exists the radial limit

$$\lim_{r \to 1} f(re^{i\theta}).$$

Solution. □

Problem 5 (2025 J-I-4)**.** Find all entire functions f and g satisfying the equation

$$e^f + e^g = 4.$$

Solution. Since e^f is never zero, e^g can never be equal to 4, and vice versa. By Picard's theorem this means e^f and e^g, hence f and g, are both constant (they are entire and miss the values 0 and 4). Thus the only functions satisfying this equation are $f \equiv a$, $g \equiv b$ for any pair of consants $a, b \in \mathbb{C}$ satisfying $e^a + e^b = 0$. □

Problem 6 (2025 J-II-1)**.** Let $f \colon \mathbb{D} \to \mathbb{D}$ be a nonconstant holomorphic map such that for every compact subset $K \subset \mathbb{D}$, the set $f^{-1}(K)$ is compact. Show that there is a number $N \in \mathbb{N}$ such that for every $a \in \mathbb{D}$, the set $f^{-1}(a)$ contains at most N distinct points in $\mathbb{D}$.

Solution. Since f is proper, $f^{-1}(a)$ is finite for each $a \in \mathbb{D}$, say with n_a elements. Then $f(z) - f(a)$ has n zeroes (with multiplicity). □

Problem 7 (2024 J-I-3)**.** Suppose f is continuous on the plane and holomorphic on $\mathbb{C} \backslash \mathbb{R}$. Prove that f is holomorphic on the whole plane.

Solution. Morera □

Problem 8 (2023 A-II-6)**.** Find the number of solutions (counting multiplicity) to $z^8 - 5z^6 + 2z^3 - z - 1 = 0$ that lie inside the unit disk.

Solution. If z is a solution to the equation with $|z| = 1$,

$$|5z^6| = |z^8 + 2z^3 - z - 1| \leq |z^8| + |2z^3| + |z| + |1| = 5 = |5z^6|,$$

so equality holds in the triangle inequality. This happens if and only if 1, z, $2z^3$, and z^8 are real multiples of one another, i.e. z is one of ± 1, but $z = 1$ and $z = -1$ are easily verified to not be solutions. Thus,

$$|z^8 + 2z^3 - z - 1| < |5z^6|$$

so Rouché's theorem implies that the given equation has the same number of zeroes in the unit disk as $5z^6$, namely $\boxed{6}$. □

Problem 9 (2023 A-I-6)**.** Let $\mathbb{D} \subset \mathbb{C}$ be the open unit disk. Is there a holomorphic function f with $f(\mathbb{D}) = \mathbb{D}$, $f(0) = f'(0) = 2/3$? If so, give a formula. If not, prove that it cannot exist.

Solution. □

Problem 10 (2023 J-I-2)**.** Set $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ and let $f : \mathbb{D} \to \{w \in \mathbb{C} : e^{-\pi/2} < |w| < e^{\pi/2}\}$ be a holomorphic map satisfying $f(0) = 1$. Show that $|f'(0)| \leq 2$.

Solution. □

Problem 11 (2022 A-I-5)**.** If $\Omega \subset \mathbb{C}$ is simply connected and $f : \Omega \to \mathbb{C}\backslash\{0\}$ is a holomorphic function, is there a holomorphic function $g : \Omega \to \mathbb{C}$ such that $f = \exp(g)$ on Ω? Prove or give a counterexample.

Solution. □

Problem 12 (2022 A-II-3)**.** Let Ω be the half-disk $\{z = x + iy : x > 0, |z| < 1\}$. Let u be the bounded harmonic function on Ω that has boundary value 1 on the vertical open segment $(-i, i)$ and has boundary value 0 on the open circular arc $\{|z| = 1, x > 0\}$. What is the value of $u(1/2)$?

Solution. □

Problem 13 (2022 J-I-1)**.** Suppose $f : \mathbb{D} \longrightarrow \mathbb{C}$ is a holomorphic function on the unit disk so that

$$f\left(\frac{1}{n}\right) = \frac{5}{n^2} \quad \forall n \in \mathbb{Z}, n \geq 2.$$

Find $f''(0)$.

Solution. By continuity we have that $f(0) = 0$, so the zero set of $f(z) - 5z^2$ is not discrete, meaning that $f(z) - 5z^2$ is the zero function. Thus $f''(z) = 10$ for all z, especially $z = 0$. □

Problem 14 (2022 J-II-4)**.** Let $U \subset \mathbb{C}$ be an open subset. Suppose $f_i : U \longrightarrow \mathbb{C}$ is a sequence of holomorphic functions converging uniformly on compact subsets to a function $f : U \longrightarrow \mathbb{C}$. Show that f is also holomorphic. Justify each step clearly.

Solution. This problem is the same as 2026 J-II-4. □

Problem 15 (2021 A-I-2)**.** Let $P(z) = z^n + a_1 z^{n-1} + \cdots + a_n$. Prove that either there is $|z| = 1$ such that $|P(z)| > 1$ or $P(z) = z^n$.

Solution. This problem is the same as 2025 A-I-3. □

Problem 16 (2021 J-I-6). By Riemann's mapping theorem there is a conformal map $f : (-1,1)^2 \to D$ from the open square into the open disc $D = \{z : |z| < 1\}$ in the complex plane which is uniquely determined by the conditions $f(0) = 0$ and $f'(0) > 1$. Show that the radius of convergence of the Taylor series for f at the origin is less than or equal to 2.

Solution. □

Problem 17 (2020 A-I-5). Let Ω be the half-disk $\{z = x + iy : x > 0, |z| < 1\}$. Let u be the bounded harmonic function on Ω that has boundary values 1 on the vertical open segment $(-i, i)$ and has boundary values 0 on the open circular arc $\{|z| = 1, x > 0\}$. What is the value of $u(1/2)$?

Solution. □

Problem 18 (2020 A-II-1). How many roots (counted with multiplicity) does the function

$$g(z) = 6z^3 + e^z + 1$$

have in the unit disk $|z| < 1$?

Solution. Since

$$|e^z + 1| \leq e + 1 < 6 = |6z^3|$$

Rouché's theorem says that $g(z)$ and $6z^3$ have the same number of solutions on the unit disk, namely $\boxed{3}$. □

Problem 19 (2020 J-II-3). Let $\mathcal{H} = \{x + iy \mid y > 0\} \subset \mathbb{C}$ be the upper half-plane, and let $\mathcal{U} = \mathcal{H} - \gamma$, where γ is the line segment $\{3 + iy \mid 0 < y \leq 1\}$. Construct an explicit holomorphic diffeomorphism $\Phi : \mathcal{U} \to \mathcal{D}$, where $\mathcal{D}$ is the open unit disk. You may leave your answer expressed as a composition of other maps.

Solution. □

Problem 20 (2019 J-I-2). Evaluate the integral

$$\int_0^\infty \frac{x^{a-1}}{x+1} dx$$

where $0 < a < 1$.

Solution. □

Problem 21 (2019 J-II-3). Denote by D_ρ the disk of radius ρ with center at 0 on the complex plane $\mathbb{C}$. For each $R > 0$ and each $p \in D_R$ let $S_R(p) = \sup\{r > 0 : \exists f \colon D_r \to D_R \text{ holomorphic}, f(0) = p, |f'(0)| = 1\}$. Find $S_R(p)$.

Solution. □

Problem 22 (2019 A-II-1). Suppose $f : \mathbb{C} \to \mathbb{C}$ is continuous on the closed disk $\{z : |z| \leq 1\} \subset \mathbb{C}$ and holomorphic on its interior. If f restricted to the boundary of this disk only takes values on the curve $xy = 1$ (where a point $z \in \mathbb{C}$ is written as $z = x + iy$, with $x, y \in \mathbb{R}$), show that f is constant on the whole disk.

Solution. □

Problem 23 (2018 J-I-2). Use the method of residues to compute the improper integral

$$\int_{-\infty}^\infty \frac{x+1}{x^4 + 2x^2 + 5} dx$$

Solution. Let $\pm\alpha$ and $\pm\beta$ be the roots of the denominator, which factors as

$$(x^2+1)^2+4=(x^2+2i+1)(x^2-2i+1),$$

and suppose also $\alpha^2=-2i-1$ and $\beta^2=2i-1$ and that α and β have positive imaginary parts (relabel and/or α and β otherwise). ☐

Problem 24 (2018 J-II-4)**.** Let $\{f_n : n=1,2,3,\ldots\}$ and f denote holomorphic functions defined on an open domain D in the complex plane. If the sequence $\{f_n\}$ converges uniformly on compact subsets of D to f, then show that the sequence of derivatives $\{f_n'\}$ converges uniformly on compact subsets of D to the derivative f'.

Solution. This problem is the same as 2026 J-II-4. ☐

Problem 25 (2018 A-I-3)**.** Show that if $c>1$, then the function

$$f(z)=ze^{c-z}-1$$

has precisely one root in $\Delta=\{|z|<1\}$, and this root is real and positive.

Solution. ☐

Problem 26 (2018 A-II-4)**.** Fix an infinite sequence $\{z_n\}$ of complex numbers in the unit disk D and suppose that for every other sequence $\{w_n\}$ in D, there is a bounded holomorphic function f on D so that $f(z_n)=w_n$ for all n. Prove that there is a constant $C<\infty$, so that we can always choose such an f that also satisfies

$$\sup_{z\in D}|f(z)|\le C\sup_n |w_n|\,.$$

Solution. ☐

Problem 27 (2017 J-I-5)**.** Is the ring of holomorphic functions on $\mathbb{C}$ a unique factorization domain? Justify your answer.

Solution. The answer is $\boxed{\text{no}}$.

Let f be an entire function which is irreducible in the ring $\mathcal{O}(\mathbb{C})$ of entire functions. If f is nowhere vanishing then it is a unit, and if f has at least two zeros (counting with multiplicity), say z_1 and z_2, then $f(z)=(z-z_1)(z-z_2)g(z)$ for some entire function g, and $z-z_1$ and $(z-z_2)g(z)$ are non-units, so f is reducible. Conversely if f has exactly one zero z_1, then if $f=gh$ for non-units g and h then g and h both have a zero, hence both are zero at z_1. This implies that gh has a zero at z_1 of order at least 2, which contradicts the fact that f has exactly one zero; thus f is irreducible. That is, the irreducible elements of $\mathcal{O}(\mathbb{C})$ are the entire functions with exactly one zero.

Now observe that $\sin z\in\mathcal{O}(\mathbb{C})$ has infinitely many zeroes while any finite product of irreducible elements of $\mathcal{O}(\mathbb{C})$ has finitely many zeroes, so $\sin z$ does not have a factorization into irreducibles, hence $\mathcal{O}(\mathbb{C})$ is not a UFD. ☐

Problem 28 (2017 J-II-5)**.** Let $\mathbb{D}\subset\mathbb{C}$ be the open unit disk and $\mathbb{H}:=\{z\in\mathbb{C}:\operatorname{Im}(z)>0\}$ the upper half plane. Let Γ be the set of holomorphic maps $f:\mathbb{D}\longrightarrow\mathbb{H}$ satisfying $f(0)=i$. Compute

$$\sup_{f\in\Gamma}\operatorname{Im}(f(3/4))$$

where $\operatorname{Im}(z)$ is the imaginary part of z for all $z\in\mathbb{C}$.

Solution. ☐

Problem 29 (2017 A-I-4)**.** Use the Residue Theorem to evaluate the integral $\int_0^\infty \frac{x^{1/2}dx}{x^3+1}$.

Solution. ☐

Problem 30 (2017 A-I-6)**.** Let $D \subset \mathbb{C}$ denote the open unit disk, and let $f : D \to \mathbb{R}$ be a harmonic function which belongs to $L^2(D)$. Prove that there is a positive constant K such that

$$|f(z)| < \frac{K}{1 - |z|}$$

for all $z \in D$.

Solution. □

Problem 31 (2016 J-I-4)**.** Let $\mathbb{D} = \{z : |z| < 1\}$ be the unit disk in $\mathbb{C}$, and $f : \mathbb{D} \to \mathbb{D}$ be a holomorphic map such that $f'(0) = 0$. Show that

$$|f(z) - f(-z)| \leq 2|z|^3$$

Solution. Define $g \colon \mathbb{D} \to \mathbb{D}$ by

$$g(z) = \begin{cases} (f(z) - f(-z))/2 & \text{if } z \neq 0 \\ 0 & \text{if } z = 0, \end{cases}$$

□

Problem 32 (2016 A-I-4)**.** Let $p : \mathbb{C} \longrightarrow \mathbb{C}$ be a polynomial with simple roots, none of which lies on the unit circle. Show that the number of roots inside the unit disk is equal to the degree of the map:

$$\rho : S^1 \longrightarrow S^1, \quad \rho\left(e^{i\theta}\right) = \frac{p\left(e^{i\theta}\right)}{\left|p\left(e^{i\theta}\right)\right|}$$

Solution. □

Problem 33 (2016 A-I-6)**.** Let Ω be a connected open subset of $\mathbb{C}$, and let F be a meromorphic function on Ω. Show that there is a meromorphic function h on Ω such that $h'(z) = F(z)$ if and only if all the residues of F vanish.

Solution. □

Problem 34 (2016 A-II-5)**.** Find all injective holomorphic maps $F : \mathbb{C} - \{0\} \longrightarrow \mathbb{C} - \{0\}$.

Solution. The solution set is $\boxed{F(z) = cz}$ and $\boxed{F(z) = \frac{c}{z}}$ for some constant c.

If F has an essential singularity at 0, then $\{f(z) : |z| < \varepsilon\}$ is dense in $\mathbb{C}$ for every $\varepsilon > 0$ by the Casorati-Weierstrass theorem so [why?] F cannot be injective. Thus 0 is either removable or a pole; similarly, by considering $F(\frac{1}{z})$, which is also an injective holomorphic map $\mathbb{C} - \{0\} \to \mathbb{C} - \{0\}$ because $z \mapsto \frac{1}{z}$ is a bijection on $\mathbb{C} - \{0\}$, we can conclude that ∞ is either removable or a pole.

Thus, F extends to a nonconstant map on the Riemann sphere. Such an extension must be surjective because the image of F is open by the open mapping theorem and closed because it is (the image of) a compact set, which means that $F(\{0, \infty\}) = \{0, \infty\}$. By replacing F with $\frac{1}{F}$ if necessary we can assume that $F(\infty) = \infty$, so that F is entire with a pole at ∞, hence a polynomial. By the fundamental theorem of algebra such a polynomial must have degree 1 (else there is a point in $\mathbb{C} - \{0\}$ whose preimage has $\deg F > 1$ points), so if $F(\infty) = \infty$ and $F(0) = 0$ then $F(z) = cz$ for some constant c. The other case of $F(\infty) = 0$ and $F(0) = \infty$ was reduced to the prior one by reciprocration, so the possible F in this case are $F(z) = \frac{c}{z}$ for some constant c. □

Problem 35 (2015 J-I-1)**.** Suppose f and g are two entire holomorphic functions with the property that $|f(z)| \leq |g(z)|$ for every $z \in \mathbb{C}$. Prove that there is a constant $\lambda \in \mathbb{C}$ such that $f = \lambda g$.

Solution. If z_0 is a zero of $g(z)$ with order d, then $g(z) = (z - z_0)^d g_0(z)$ for some entire function g_0 which is not zero at z_0. The given inequality implies z_0 is a zero of $f(z)$ as well, so write $f(z) = (z - z_0)^{d'} f_0(z)$ for some entire function f_0 which is not zero at z_0. If $d' < d$, then

$$|f_0(z)| \leq |(z - z_0)^{d-d'} g_0(z)|,$$

which implies $f_0(z)$ is zero at z_0; this is a contradiction, so $d' \geq d$. That is, every zero of g is a zero of f with higher order in f than in g, so the singularities of f/g at the zeroes of g are all removable. In other words f/g extends to an entire function which by the given information satisfies $|f(z)/g(z)| \leq 1$ for all z, whence Liouville's theorem implies $f/g = \lambda$ for some constant $\lambda \in \mathbb{C}$. □

Problem 36 (2015 A-I-3)**.** Let f be a holomorphic function defined on a neighborhood of the closed unit disk $\overline{\mathbb{D}} := \{z \in \mathbb{C} : |z| \leq 1\}$, and assume that $|f(z)| < 1$ for all $|z| = 1$. Determine the number of fixed points of f in $\overline{\mathbb{D}}$.

Solution. □

Problem 37 (2015 A-I-5)**.** Prove that if $\{u_n\}$ is a sequence of harmonic functions on the unit disk with $|u_n(x)| \leq 1$ for all x and all n, then there is a subsequence which converges at every point of unit disk.

Solution. □

Problem 38 (2014 J-I-4)**.** Use the method of residues to compute the improper integral $\int_0^\infty \frac{x^2}{x^6+x^4+x^2+1} dx$

Solution. □

Problem 39 (2014 J-II-1)**.** Set $f(z) = \sum_{n=1}^\infty f_n(z)$, where $f_n(z) = \frac{z^n}{10^{n^2}}$. Show that $f(z)$ is an entire function which has exactly n distinct zeros inside of each circle $C_n = \{z : |z| = 100^n\}, n = 1, 2, 3, 4, \ldots$. (Hint: compare the functions $|f_n(z)|$ and $|f(z) - f_n(z)|$ on C_n.)

Solution. □

Problem 40 (2014 A-I-5)**.** Let $\mathbb{D} \subset \mathbb{C}$ be the unit disk, $f : \mathbb{D} \to \mathbb{D}$ a holomorphic function. Suppose that $f(c) = c$ for some $c \in \mathbb{D}$, and $|f'(c)| < 1$. Prove that for any point $z_0 \in \mathbb{D}$, the sequence $\{z_n\}$ defined by iteration, $z_n = f(z_{n-1})$, converges to c. (Hint: first consider the case $c = 0$.)

Solution. □

Problem 41 (2014 A-II-2)**.** Let $\Omega \subset \mathbb{C}$ be the right half-plane with the disk D removed, where D is the disk of radius $r = 3$ centered at $w_0 = 5$. What is the maximum number of disjoint open disks in Ω whose boundaries touch both boundary components of Ω ? Prove your answer. (Hint: transform the domain.)

Solution. □

Problem 42 (2013 J-I-1)**.** What is the average of the angle $\alpha(z)$ subtended by the segment $[0, 1]$ at a point z in the disk $D = \{z \in \mathbb{C} : |z - i| \leq 1\}$? Hint: express $\alpha(z)$ in terms of a holomorphic function on D.

Solution. □

Problem 43 (2013 J-II-2)**.** Show that the critical points of a complex polynomial $p(z)$ all lie in the closed convex hull of its roots. For the sake of simplicity, you may assume that the roots of $p(z)$ are all distinct.

Solution. □

Problem 44 (2013 A-I-3)**.** Suppose $f : D(0, 1) \to D(0, 1)$ is a holomorphic map from the open unit disc into the closed unit disc. Can the image of f contain both the points 0 and 1 ?

Solution. □

Problem 45 (2013 A-II-1)**.** Write an explicit formula for the conformal map from a complex z-plane Δ with removed rays $-\infty < x < -1, 1 < x < \infty$ to a strip $\Omega = \{0 < \operatorname{Im} w < 7\}$. The map transforms the left ray to the bottom boundary component of Ω and the right ray to the top boundary component respectively.

Solution. □

Problem 46 (2012 J-I-3)**.** Let $p(z)$ be a complex polynomial whose critical values are all in the open, complex, unit disk Δ, i.e., $p'(z) = 0$ implies $p(z) \in \Delta$. Prove that $p^{-1}(\Delta)$ is connected.

Solution. □

Problem 47 (2012 J-II-2)**.** Let f be a continuous, complex-valued function on the closed disk $\bar{\Delta} = \{z \in \mathbb{C} : |z| \leq 1\}$ which is holomorphic on the interior. Assume that f restricted to the boundary $\partial\bar{\Delta}$ only takes values on the curve $\{x + iy \in \mathbb{C} \mid x, y \in \mathbb{R}, xy = 1\}$. Show that f is constant on the whole disk.

Solution. □

Problem 48 (2012 A-I-4)**.** Find, with proof, the precise number of zeroes of the complex polynomial $p(z) = z^9 - 2z^6 + z^2 - 8z - 2$ inside the annulus $1 < |z| < 2$.

Solution. □

Problem 49 (2012 A-II-6)**.** Denote by $(\mathcal{H}, \langle \bullet, \bullet \rangle)$ the Hilbert space of holomorphic functions on the unit disk $D \subset \mathbb{C}$ which is the closure of the set of complex polynomial functions on the disk with respect to the Hermitian inner product

$$\langle f(z), g(z) \rangle = \int_0^{2\pi} f\left(e^{i\theta}\right) \overline{g\left(e^{i\theta}\right)} \frac{d\theta}{2\pi}.$$

Prove that the span of $\{1/(z-n)\}_{n=2}^{\infty}$ is dense in $\mathcal{H}$.

Solution. □

Problem 50 (2011 J-I-1)**.** Prove that any three distinct points in the Euclidean plane $\mathbb{R}^2$ are the vertices of a topological triangle whose edges are arcs of Euclidean circles, such that the interior angle at each vertex is $90°$.

Solution. □

Problem 51 (2011 J-I-4)**.** Let $u : \mathbb{C} \to \mathbb{R}$ be a $\mathscr{C}^2$-smooth, subharmonic function. Show that the function $\Psi : \mathbb{C} \to \mathbb{R}$ defined by

$$\Psi(z) := \frac{1}{2\pi} \int_0^{2\pi} u\left(e^z e^{i\theta}\right) d\theta$$

is subharmonic. Then show that the function $\mathbb{R} \ni x \mapsto \Psi(x + i0)$ is convex and increasing.

Solution. □

Problem 52 (2011 A-I-3)**.** Let $\Omega \subset \mathbb{C}$ be an convex open set, and let $f : \Omega \to \mathbb{C}$ be a holomorphic function such that $\operatorname{Re} f'(z) > 0$ for all $z \in \Omega$. Prove that f is injective.

Solution. □

Problem 53 (2011 A-II-3)**.** If $p(z) = z^n + a_1 z^{n-1} + \cdots + a_n$ is a monic polynomial, prove that

$$\sup\{|p(z)| : |z| \leq 1\} \geq 1.$$

Solution. This problem is the same as 2025 A-I-3. □

Problem 54 (2010 J-I-1)**.** Let $f : \mathbb{C} - \{0\} \to \mathbb{C}$ be a holomorphic function on the punctured complex plane, and suppose that $f(2z) = f(z)$ for all $z \neq 0$. Prove that f is constant.

Solution. The annulus $A = \{z \in \mathbb{C} : 1 \leq |z| \leq 2\}$ is compact, so $f(A)$ is compact. Then, if $z \in \mathbb{C}$ has magnitude $r > 0$, let n be the integer such that $1 \leq 2^n r < 2$ (precisely, $n = -\log_2 r$), so $2^n z \in A$. We are given that $|f(z)| = |f(2^n z)|$, so $f(z) \in A$ for all $z \in \mathbb{C}$. Thus the image of f is a compact (hence closed) subset of $\mathbb{C}$. If f is nonconstant then its image is also open by the open mapping theorem (since the domain $\mathbb{C} - \{0\}$ is open in $\mathbb{C}$), which implies that the image is $\mathbb{C}$ because $\mathbb{C}$ is connected — but this is absurd since $\mathbb{C}$ is not compact. □

Problem 55 (2010 J-II-6)**.** A fractional linear transformation maps the annulus $r < |z| < 1$(where $r > 0$) onto the domain bounded by the two circles $\left|z - \frac{1}{4}\right| = \frac{1}{4}$ and $|z| = 1$. Find r.

Solution. □

Problem 56 (2010 A-I-4)**.** Prove that an entire function $f(z)$ on the complex plane is a polynomial if $|f(z^2)| \leq |f(z)|^2$ for every z.

Solution. □

Problem 57 (2010 A-II-3)**.** Let $f : \Delta \to \mathbb{C}$ be a holomorphic function on the open unit disk. Decompose f into its real and imaginary parts, $f = u + iv$. Among all holomorphic functions f with $v(0) = 0$ and with $|u| < 1$, what is the largest possible value of $v(1/2)$?

Solution. □

Problem 58 (2009 J-I-3)**.** Let $S = [-1, 1]$ denote the closed line segment from -1 to 1 and suppose f is the Riemann map from the unit disk to the domain

$$\Omega = \{z : \operatorname{dist}(z, S) < 1\}.$$

What is the smoothness of $\partial\Omega$ and what is the radius of convergence of the power series of f around the origin?

Solution. □

Problem 59 (2009 J-I-4)**.** Suppose that u is a harmonic function on all of $\mathbb{R}^n$. Show that if $u \in L^1(\mathbb{R}^n)$, then $u \equiv 0$. Say $1 < p < \infty$. Show that if $u \in L^p(\mathbb{R}^n)$, then $u \equiv 0$.

Solution. □

Problem 60 (2009 A-I-1)**.** Let $p(z)$ and $q(z)$ be two polynomials over the complex numbers $\mathbb{C}$, and suppose that their degrees satisfy $\deg(p) \leq \deg(q) - 2$. Let $C_r \subset \mathbb{C}$ be the circle radius r centered at 0 , and assume that each root ζ of $q(z)$ satisfies $|\zeta| < r$. Compute the contour integral

$$\int_{C_r} \frac{p(z)}{q(z)} dz$$

Solution. □

Problem 61 (2009 A-II-1)**.** Let $u : \mathbb{C} \to \mathbb{R}$ be a non-constant harmonic function on the complex plane. Show that the subset $\{z \in \mathbb{C} \mid u(z) = 0\}$ is unbounded.

Solution. □

Problem 62 (2009 A-II-5)**.** Let $\Omega = \mathbb{C} \backslash \{a, b, c\}$ be the complement of three distinct points in the complex plane. Show that the set of one-to-one holomorphic maps $f : \Omega \to \Omega$ forms a finite group. Is the order of this group independent of the choice of $\{a, b, c\}$?

Solution. □

Problem 63 (2008 J-I-6)**.** Let Ω denote the open subset $\{x + iy \mid -1 < y < 1\}$ of the complex plane, and let $f : \Omega \longrightarrow \Omega$ denote a holomorphic function which satisfies $f(-1) = -1$ and $f(1) = 1$. Show that f is the identity map.

Solution. □

Problem 64 (2008 J-II-3)**.** Compute the real integral $\int_{-\infty}^{\infty} \frac{1}{1+x^4} dx$.

Solution. □

Problem 65 (2008 A-I-1)**.** Find all entire functions f such that $|f'| \leq |f|$ everywhere.

Solution. □

Problem 66 (2008 A-II-2)**.** Let Ω be the right half-plane ($\{x + iy : x > 0\}$) with the disk of radius $r < 1$ centered at 1 removed. Let $N(r)$ be the maximum number of disjoint open disks in Ω whose boundaries touch both boundary components of Ω. Show

$$N(r) = \left\lfloor \frac{\pi}{\arcsin\left(\sqrt{\frac{1-r}{1+r}}\right)} \right\rfloor .$$

(Here $\lfloor \cdot \rfloor$ denotes the greatest integer function.)

Solution. □

Problem 67 (2007 J-I-2)**.** Find an explicit holomorphic bijection $\zeta = f(z)$ between the crescent

$$\{z \in \mathbb{C} : |z| \leq 1,\ |z + 1| \geq \sqrt{2}\}$$

and the disk

$$\{\zeta \in \mathbb{C} : |\zeta| \leq 1\}.$$

Solution. □

Problem 68 (2007 J-II-1)**.** Let $f(z)$ be a continuous complex-valued function on the closed unit disk $D = \{z \in \mathbb{C} || z |\leq 1\}$. If f is holomorphic in the interior $|z| < 1$ of D, and real-valued on the boundary circle $|z| = 1$, show that f must be constant.

Solution. □

Problem 69 (2007 A-I-6)**.** Let us consider a rational function $f(z) = z\frac{z-a}{1-\bar{a}z}$ of complex variable, where $|a| < 1$. Show that f restricts to a degree two covering map from the unit circle $\mathbb{T}$ to itself. Show that the Lebesgue measure μ on $\mathbb{T}$ is invariant under f, i.e.,

$$\int \phi \circ f \, d\mu = \int \phi \, d\mu$$

for any continuous function ϕ on $\mathbb{T}$. (Hint: make use of the Dirichlet problem in the unit disk.)

Solution. □

Problem 70 (2007 A-II-1)**.** Let $\mathbf{D} = \{z : |z| < 1\}$ be the unit disk, and let $f : \mathbf{D} \to \mathbf{D}$ be a holomorphic function such that $f(0) = \alpha \neq 0$. Prove that f does not have zeros in the disk $\{z : |z| < |\alpha|\}$.

Solution. □

Problem 71 (2006 J-I-2)**.** Let

$$f(z) = \sum_{n=0}^{\infty} z^{2^n}, \quad |z| < 1$$

Prove that for every ζ satisfying $\zeta^{2^l} = 1$ for some $l \in \mathbb{N}$,

$$\lim_{r \to 1^-} f(r\zeta) = \infty.$$

Solution. □

Problem 72 (2006 J-I-4)**.** Let $p > 2$ be a prime number. Find the absolute value of

$$S = \sum_{n=0}^{p-1} \exp\left\{\frac{2\pi i n^2}{p}\right\}.$$

Solution. □

Problem 73 (2006 J-II-2)**.** Show that there is no meromorphic function satisfying

$$f(z+1) = f(z), \quad f(z+i) = f(z)$$

for all $z \in \mathbb{C}$, whose only poles are simple poles at the points $m + ni$, $m, n \in \mathbb{Z}$.

Solution. □

Problem 74 (2006 A-I-4)**.** Let $G = \{z \in \mathbb{C} \mid \text{Re}z > 0 \text{ and } \text{Im}z > 0\}$, and let $f : G \to G$ be a non-trivial conformal map such that $f \circ f = id$. Prove that f has precisely one fixed point inside G.

Solution. □

Problem 75 (2006 A-II-1)**.** Find

$$\lim_{n \to \infty} \int_0^{\infty} \left(\frac{x}{\sqrt{n}}\right)^{n^2} e^{-nx} dx$$

Solution. □

Problem 76 (2006 A-II-4)**.** Show that $z^7 - 4z^3 + z - 1$ has 3 zeros inside the unit circle (counted with multiplicity).

Solution. □

Problem 77 (2005 J-I-2)**.** Suppose T is an equilateral triangle of side length 1 and let $f_- : T \to D = \{z : |z| < 1\}$ be a conformal map which sends the center c of the triangle to the origin in the disk (this is a 1-to-1, onto holomorphic map between the interiors which you may assume extends continuously to the boundary). What is the radius of convergence of the power series of f around the point c ?

Solution. □

Problem 78 (2005 A-II-1)**.** Let $\Omega = \mathbb{C}\backslash[-1, 1]$. Give an example of a non-constant, bounded holomorphic function on Ω. Show that such a function cannot extend continuously to all of $\mathbb{C}$.

Solution. □

Problem 79 (2004 J-I-3)**.** Suppose $\Omega = \{z \in \mathbb{C} : |z - 1| > 1 \text{ and } |z + 1| > 1\}$. Let U be a harmonic function on Ω with boundary values $U = 0$ on $\{|z - 1| = 1\}$ and $U = 1$ on $\{|z + 1| = 1, z \neq 0\}$. Find the value of $U(4)$.

Solution. □

Problem 80 (2004 J-I-6)**.** Let g denote a holomorphic function on subset of the complex plane given by $|z| < r$, were r is a fixed real number satisfying $r > 1$. Suppose that $g(z) \leq 1$ holds for all $|z| \leq 1$. Show that for all $t \in \mathbb{C}$ with $|t| < 1$, the equation

$$z = tg(z)$$

has a unique solution $z = s(t)$ in the disc$|z| < 1$. Show that $t \to s(t)$ is a holomorphic function on the disc$|t| < 1$.

Solution. □

Problem 81 (2004 A-I-3)**.** Suppose f is a holomorphic function in the open unit disc $D = \{z \in \mathbb{C} || z |< 1\}$ such that $f(0) = 0$ and $|f(z) + zf'(z)| < 1$ for all $z \in D$. Show that $|f(z)| \leq |z|/2$ for all $z \in D$.

Solution. □

Problem 82 (2004 A-II-3)**.** Let $f, g \in \mathcal{O}(D) \cap C(\bar{D})$ where $D \subset \mathbb{C}$ is the open unit disc $\{z \in \mathbb{C} \mid |z| < 1\}$. Suppose there is a circle homeomorphism $h : \partial D \longrightarrow \partial D$ such that $f(t) = (g \circ h)(t)$ for all $t \in \partial D$. Prove that $f(\bar{D}) = g(\bar{D})$; i.e. the two functions have the same range on the closed disc.

Solution. □

Problem 83 (2003 J-I-1)**.** Let f be a holomorphic function on the punctured plane $0 < |z| < \infty$. Assume that there exists a positive constant C and a real constant N such that

$$|f(z)| \leq C|z|^N \text{ for } 0 < |z| < \frac{1}{2}.$$

Show that $z = 0$ is either a pole or a removable singularity for f and find a bound for ord $\mathrm{ro}_0 f$, the order of the function f at 0 .

Solution. □

Problem 84 (2003 J-I-2)**.** State any form of Picard's Theorem. Let $n \geq 2$ be an integer. Show that there are no nowhere vanishing and nonconstant entire functions f and g that satisfy

$$f^n + g^n = 1.$$

Assume now that $n > 2$. What are all the solutions f and g in the ring of entire functions that satisfy

$$f^n + g^n = 1?$$

Hint: transform the problem to the setting of meromorphic functions on the complex line.

Solution. □

Problem 85 (2003 J-II-1)**.** Suppose f is a meromorphic function on the plane which is holomorphic on the unit disk and has one simple pole only on the unit circle at the point z_0. Let $\sum_{n=0}^{\infty} a_n z^n$ be the power series for f at the origin. Show that $\lim_{n\to\infty} a_n z_0^n$ exists.

Solution. □

Problem 86 (2003 A-I-1)**.** Prove that for any $r > 0$ there exists $N > 0$ such that for every $n > N$, the polynomials

$$P_n(z) = 1 + z + \frac{z^2}{2!} + \ldots + \frac{z^n}{n!}$$

have no zeroes in the disk $\{z \in \mathbf{C} : |z| < r\}$.

Solution. □

Problem 87 (2003 A-I-2)**.** Let f be a holomorphic function on the unit disk $\{z \in \mathbf{C} : |z| < 1\}$ such that $|f(z)| < 1$ for $|z| < 1$ If $f\left(\frac{1}{2}\right) = \frac{1}{3}$, find a sharp upper bound for $\left|f'\left(\frac{1}{2}\right)\right|$.

Solution. □

Problem 88 (2003 A-II-1)**.** Show that

$$\lim_{M\to\infty}\int_{S_M}\frac{e^{iz}}{z}dz = 0,$$

where S_M is the semicircle in the upper half plane with center at 0 and radius $M > 0$. Evaluate

$$\int_0^\infty \frac{\sin x}{x}dx.$$

Solution. □

Problem 89 (2002 J-I-1)**.** Suppose Ω is a horizontal strip $\{z : \mathrm{Im}z \in (a,b)\}$. Suppose f is holomorphic in Ω and $f(z+1) = f(z)$ for all $z \in \Omega$. Prove that f has an expansion

$$f(z) = \sum_{-\infty}^{\infty} c_n e^{2\pi inz}$$

which converges uniformly in $\{\mathrm{Im}z \in (a+\epsilon, b-\epsilon)\}$ for all $\epsilon > 0$.

Solution. □

Problem 90 (2002 J-I-4)**.** Let $\{a_n\}_{n=0}^\infty$ be a sequence of complex numbers tending toward 0 . By definition, the infinite product $\prod_{n=0}^\infty (1+a_n)$ converges iff $\sum_{n=0}^\infty \log(1+a_n)$ converges to a finite value. The product is said to converge absolutely if the corresponding sum does. Show that $\prod_{n=0}^\infty (1+a_n)$ is absolutely convergent iff $\sum |a_n| < \infty$. Show that for $\mathrm{Re}(s) > 1$, the infinite product $\prod_{p \text{ prime}} \frac{1}{1-p^{-s}}$ is absolutely convergent. For $\mathrm{Re}(s) > 1$, let $\zeta(s) = \sum_{n=1}^\infty n^{-s}$. Show that the infinite product from (ii) equals $\zeta(s)$ whenever $\mathrm{Re}(s) > 1$. (Hint: expand $\left(1-p^{-s}\right)^{-1}$ in a geometric series for each p.)

Solution. □

Problem 91 (2002 J-II-2)**.** Show that there is no entire function $f(z)$ with the property that, for each point w in the plane, $f^{-1}(w)$ consists of exactly two points.

Solution. □

Problem 92 (2002 A-I-5)**.** Let D be the domain in the extended complex plane $C \cup \{\infty\}$ exterior to the circles $|z-1| = 1$ and $|z+1| = 1$. Find a Riemann map (one-to-one holomorphic map) of D onto the strip $S = \{z \in C; 0 < \mathrm{Im}z < 2\}$.

Solution. □

Problem 93 (2002 A-II-2)**.** Let II be the upper half-plane, and $f : \mathbb{E} \to \mathbb{H}$ an antiholomorphic map such that $f \circ f = I$ d Show that f can be written as $f(z) = \overline{g(z)}$ where

$$g(z) = (az+b)/(cz+d), \quad a,b,c,d \text{ real, } ad-bc = -1$$

Show that f is conjugate to $f_0 : z \mapsto -\bar{z}$, i.e.,

$$f = \phi \circ f_0 \circ \phi^{-1}$$

for some biholomorphic map $\phi : II \to \text{II}$

Solution. □

Problem 94 (2002 A-II-4)**.** Let Ω be a connected and simply-connected region in $\mathbb{C}$ Show that for any real-valued function u which is harmonic in Ω (i.e. $u \in \mathcal{C}^2(\Omega), u_{xx} + u_{yy} = 0$ in Ω, with subscripts denoting partial derivatives), there exists a real-valued function v such that $u + iv$ is bolomorphic in Ω. Give an example which shows that the conclusion of (i) can fail if Ω is not simply connected.

Solution. □

Problem 95 (2001 J-I-6)**.** Let $f(z)$ be a holomorphic function in $|z| \leq 1$ and $u = \mathrm{Re} f$. Prove the following formula: for $|z| < 1$,

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} u(\zeta) \frac{\zeta + z}{\zeta - z} d\theta + \mathrm{i}C, \quad \zeta = e^{\mathrm{i}\theta}$$

where C is some real constant.

Solution. □

Problem 96 (2001 J-II-5)**.** Let U be a domain in a complex plane such that its universal cover is the unit disk D (that is, there is a map $\pi : D \to U$ which is holomorphic and is a covering map in the usual topological sense). Show that if f is an entire function (i.e., a holomorphic function on $\mathbb{C}$) such that $f(z) \in U$ for all z, then f is constant.

Solution. □

Problem 97 (2001 J-II-6)**.** Calculate

$$\int_0^\infty \frac{\log x}{x^2 + a^2} dx$$

where $a \in \mathbb{R}, a > 0$.

Solution. □

Problem 98 (2001 A-I-2)**.** Evaluate, with proof, $\lim_{t \to 0} \frac{1}{t} \int_0^1 \sin\left(\frac{t}{\sqrt{x}}\right) dx$.

Solution. □

Problem 99 (2001 A-I-5)**.** Let $\Omega = \{z = x + iy : |z| < 1, y > 0\}$, Find an explicit 1-1, onto holomorphic map of Ω to the unit disk.

Solution. □

Problem 100 (2001 A-II-5)**.** Given a finite sequence of points $\{a_k\}$ in the open unit disk $\{|z| < 1\}$, consider the complex function

$$B(z) = \prod_{k=1}^n \frac{z - a_k}{1 - \overline{a_k} z}.$$

Show that B maps the unit circle $\{|z| = 1\}$ into itself and that if $B(0) = 0$, and g is a continuous function on the circle, then $\int_0^{2\pi} \left|g\left(e^{i\theta}\right)\right| d\theta = \int_0^{2\pi} \left|g \circ B\left(e^{i\theta}\right)\right| d\theta$, (i.e., the composition operator $\to g \circ B$ is an isometry for the L^1 norm).

Solution. □

Problem 101 (2000 J-I-4)**.** Let $f(z) = \sum_{n=1}^\infty z^{2^n}$ for $|z| < 1$. Show that $|f(z)| \leq C + C \log \frac{1}{1-|z|}$ for some $C < \infty$ and every $|z| < 1$. (Hint: choose N so that $2^{-N} \simeq 1 - |z|$ and cut the series at N.)

Solution. □

Problem 102 (2000 A-I-1)**.** Let D be the domain in the extended complex plane $\mathbb{C} \cup \{\infty\}$ exterior to the circles $|z-1|=1$ and $|z+1|=1$. Find a Riemann map (one-to-one holomorphic map) of D onto the strip $S = \{z \in \mathbb{C}; 0 < \mathrm{Im} z < 2\}$.

Solution. □

Problem 103 (2000 A-I-6)**.** Show that $\lim_{M\to\infty} \int_{S_M} \frac{e^{iz}}{z} dz = 0$, where S_M is the semi-circle in the upper half plane with center at 0 and radius $M > 0$. Evaluate $\int_0^\infty \frac{\sin x}{x} dx$.

Solution. □

Problem 104 (2000 A-II-1)**.** Let f be a holomorphic function on $|z| < 1$ such that $|f(z)| < 1$ for all $|z| < 1$ and $f\left(\frac{1}{2}\right) = \frac{1}{3}$. Find a sharp upper bound for $\left|f'\left(\frac{1}{2}\right)\right|$.

Solution. □

Problem 105 (2000 A-II-5)**.** Let $\mathbb{D}$ be the unit disk in the complex plane $\mathbb{C}$ endowed with the Lebesgue measure. Consider the Hilbert space $H = L^2(\mathbb{D})$. Show that the functions z^n are orthogonal in H. Let $f(z)$ be an analytic function in some neighborhood of the closed disk $\overline{\mathbb{D}}$. Then

$$\int_D |f(z)|^2 dxdy = \pi \sum_{n=0}^{\infty} \frac{1}{n+1} |a_n|^2 ,$$

where a_n are the Taylor coefficients of f at the origin.

Solution. □

Problem 106 (1999 J-I-3)**.** Suppose f is an entire function so that $f^{-1}(K)$ is a compact set for every compact K in the complex plane. Prove that f is a polynomial.

Solution. □

Problem 107 (1999 J-II-6)**.** Suppose $D_0 \subset \mathbb{R}^2$ is a disk centered at the origin and $\{D_k\}$, $k = 1, \ldots, n$, are disjoint closed subdisks centered on the real axis. Let u be the harmonic function in

$$\Omega = D_0 \backslash \bigcup_k D_k$$

with boundary values 1 on ∂D_0 and 0 on the ∂D_k, $k = 1, \ldots, n$. Prove that u has exactly $n-1$ critical points, and all of them are on the real axis.

Solution. □

Problem 108 (1999 A-I-3)**.** Suppose $D = \{|z| < 1\}$ is the unit disk and Ω is a simply connected domain. Suppose f and g are holomorphic maps of D into Ω such that $f(0) = g(0)$ and that f is also 1-1 and onto. Prove that $g(D(0,r)) \subset f(D(0,r))$ for every $0 < r < 1$.

Solution. □

Problem 109 (1999 A-II-5)**.** Let g be a holomorphic function on $|z| < R$, $R > 1$, with $|g(z)| < 1$ for all $|z| < 1$. Show that for all $t \in \mathbb{C}$ with $|t| < 1$, the equation

$$z = tg(z)$$

has a unique solution $z = s(t)$ in the disc $|z| < 1$. Show that $t \mapsto s(t)$ is a holomorphic function on the disc $|t| < 1$. (Hint: find an integral formula for s.)

Solution. □

Problem 110 (1998 J-I-3)**.** Let $f : S \to S$ be a conformal automorphism of the square $-3 < \mathrm{Re}z, \mathrm{Im}z < 3$ with $f(0) = 2 + 2i$. Show that
$$\frac{\sqrt{2}}{6} < |f'(0)| < \frac{10\sqrt{2}}{6}.$$

Solution. □

Problem 111 (1998 J-I-4)**.** Let f be a non-constant periodic entire function with $f(z+1) = f(z)$. Show that f has a fixed point.

Solution. □

Problem 112 (1998 J-II-4)**.** Let n be an integer, $n > 1$. Let $h : D \to \mathbb{C}$ be a harmonic function on the unit disk which has boundary values 1 on the segment $\{e^{i\theta} \mid \theta \in (0, 2\pi/n)\}$ and has boundary values 0 on the exterior of this arc. Show that $h(0) = 1/n$.

Solution. □

Problem 113 (1998 A-I-2)**.** In $\mathbb{C}$, let $D = \{z : |z| < 1\}$, $H_+ = \{z : \mathrm{Im}z > 0\}$, $H_- = \{z : \mathrm{Im}z < 0\}$, $S = \{z : z \in \mathbb{R} \text{ and } -1 < z < 1\}$. Suppose that g is continuous on the closure of D and is holomorphic in D. For $z \in \mathbb{C}\backslash S$, let
$$F(z) = \frac{1}{2\pi i}\int_{-1}^{1} (z-t)^{-1} g(t)\, dt.$$
Show that F is holomorphic in $\mathbb{C}\backslash S$. Show that F, restricted to H_+, has a holomorphic extension to $H_+ \cup D$; call that extension F_+. Similarly, F, restricted to H_-, has a holomorphic extension to $H_- \cup D$; call that extension F_-. On D, consider the function $G = F_+ - F_-$. Express G in terms of g without using any integrals.

Solution. □

Problem 114 (1998 A-I-6)**.** Let $p(z) = \sum_{k=0}^{n} a_k z^k$ be a polynomial over $\mathbb{C}$, with $a_0 = a_n = 1$. Let ω be a primitive nth root of unity, so that the nth roots of unity are $\{1, \omega, \omega^2, \ldots, \omega^{n-1}\}$. Prove that $\sum_{j=0}^{n-1} p(\omega^j) = 2n$. Suppose that whenever $0 < j < n$, $|p(\omega^j)| < 2$. Show that $p(z) = z^n + 1$ for all z.

Solution. □

Problem 115 (1998 A-II-2)**.** In $\mathbb{C}$, let $L = \{z : z \in \mathbb{R} \text{ and } z < 0\}$. Show that there exists a holomorphic function h on $\mathbb{C}\backslash L$ such that for all $z = re^{i\theta} \in \mathbb{C}\backslash L$ (with $r > 0$, $-\pi < \theta < \pi$), we have $|h(z)| = r^{\theta}$. (Hint: in polar coordinates, the Laplacian on $\mathbb{R}^2$ is $\Delta = r^{-1}\partial_r(r\partial_r) + r^{-2}\partial_\theta^2$.)

Solution. □

Problem 116 (1997 J-I-3)**.** Let n be a non-negative integer. What is the most general complex-analytic function $f : \mathbb{C} - \{0\} \to \mathbb{C}$ such that $f(2z) = 2^n f(z)$ for all $z \in \mathbb{C} - \{0\}$? Explain.

Solution. □

Problem 117 (1997 J-II-1)**.** Let $f(z)$ be a continuous complex-valued function on the closed unit disc $|z| \le 1$ which is holomorphic in the open disc $|z| < 1$. Suppose that $f(e^{i\theta}) = 0$ for $\theta \in [0, \pi/2]$. Show that $f \equiv 0$.

Solution. □

Problem 118 (1997 A-I-3)**.** Let $\Sigma \subset \mathbb{C}$ be the unit square $[0,1] \times [0,1]$, and let $P \subset \mathbb{C}$ be the rectangle $[0,1] \times [0,2]$. Is there a homeomorphism $f : \Sigma \to P$ which is complex-analytic in the interior of Σ and takes the corners of Σ to the corners of P? Explain.

Solution. □

Problem 119 (1997 A-II-4). Let U be a connected open subset of the complex plane $\mathbb{C}$, and suppose that $f : \mathbb{C} \to U$ is both an entire analytic function and a covering map. Show that the fundamental group of U is Abelian.

Solution. □

Problem 120 (1997 A-II-5). Let $z = x + iy$ be the usual complex coordinate on $\mathbb{C}$, and let $\Sigma \subset \mathbb{C}$ be the unit square $0 \leq x \leq 1$, $0 \leq y \leq 1$. Suppose that $f : \Sigma \to \mathbb{C}$ is a smooth function which satisfies $f|_{\partial\Sigma} \equiv 0$. Prove that

$$\int_\Sigma \left|\frac{\partial f}{\partial \bar{z}}\right|^2 dx\, dy \geq \frac{\pi^2}{2} \int_\Sigma |f|^2 dx\, dy.$$

When does equality occur?

Solution. □